MIL-C-36676
18 June 1970

MILITARY SPECIFICATION

CUTTER, BONE PIN, 18 inch

This specification is mandatory for use by all partments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a cutter, Steinmann type, for cutting Steinmann and Kirschner bone pins and wire.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical, General Specifications.
QQ-T-570	Tool Steel, Alloy.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.

Military

MIL-P-116	Preservation, Methods of.

FSC 6515

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STANDARDS

Federal

Test Method 151 Metals, Test Methods.

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center (DPSC)

Drawing No. 24230 CUTTER, BONE PIN, 18 INCH

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (A.N.S.I.)

A.N.S.I. B46.1 Surface Texture (Surface Roughness, Waviness and Lay)

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018.)

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3. REQUIREMENTS

3.1 General requirements. The bone pin cutter shall be in accordance with the applicable requirements of GG-I-52[illegible] and as specified herein.

3.2 Materials. Except as specified herein, materials shall conform to the applicable portions of the specifications, standards, and drawing specified and shall meet the requirements specified herein. All materials shall be of the highest quality and suitable for the purpose intended.

3.3 Hardness. Component parts of the cutter shall have hardness values in accordance with class and type of material used in its fabrication, as indicated in Table I.

Table I. Hardness Requirements

Class/Type of Material Used	Required Rockwell Hardness
Type 17-4 PH stainless	C35 to C45
Class D2 Tool steel	C59 to C64
Type 440A Martensitic stainless	C50 to C56
Type 440C Martensitic stainless	C59 to C64
Type 302 Austenitic stainless	B75 to B85
Type 303 Austenitic stainless	B75 to B85
Type 304 Austenitic stainless	B75 to B85

For small size parts (bolts, screws etc.) a minimum of two hardness determinations shall be made on each part tested from opposing sides or ends of part. Hardness values for any individual part tested shall not vary by more than four (4) points Rockwell, and also be within the limits specified.

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3.4 Style, design and dimensions. Shall be a triple action bone pin and wire cutter, Steinmann type, suitable for use in cutting Steinmann and Kirschner bone pins and wires which have a Rockwell hardness up to C45, and diameters up to and including 3/16 inch. There shall be an adjusting link on each handle of the cutter to preset or limit the angular movement of the handle. The total angle of movement of the handles between fully open and closed jaws shall not exceed 120 degrees. The jaws shall be properly aligned so that the cutting edges shall meet accurately on closure along the entire cutting edge. Jaws shall not be plated. The design construction and overall dimensions of the cutter shall be in accordance with Defense Personnel Support Center (DPSC) Drawing No. 24230.

3.5 Finish. The faces of the cutting jaws (components 5 and 6 on drawing), and both surfaces of the top and bottom aligning plates (components 7 and 8 on drawing) shall have a surface finish equal to, or superior to, an average value of 8 microinches (MU) roughness height in accordance with A.N.S.I. Standard B46.1 when tested as specified in 4.4.1. Other components of the cutter shall be finished as specified on drawing and applicable requirements of GG-I-526.

3.6 Passivation. All components fabricated from stainless steel shall be properly passivated.

3.7 Identification marking. The cutter shall be permanently marked as indicated on drawing. The word "stainless" shall be marked, except when the handles are fabricated from malleable iron.

4. QUALITY ASSURANCE PROVISIONS

4.1.1 Responsibility for inspection. Shall be in accordance with GG-I-526.

4.1.2 Inspection. Shall be in accordance with GG-I-526.

4.1.3 Certificates of quality. Certificates of quality supplied by the manufacturer of the tool steel or stainless steel may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Sampling.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and as indicated in Table II. Unit of product for examination purposes shall be one cutter.

Table II. Sampling Criteria for Examination

Type defect	Inspection level*	AQL (percent defective)
Major	S-4	1.0
Minor	S-4	2.5

* If acceptance number is not exceeded, lot shall be acceptable. If rejection number is equalled or exceeded, then the inspection at level II may be performed to determine the final rejection or acceptance.

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in Table III. Unit of product for test purposes shall be one cutter.

Table III. Sampling Criteria for Tests

Component	Characteristic	Requirement	Test procedure	Insp. level	AQL
Cutter	Composition analysis	3.2 and GG-I-526	GG-I-526	**	*
	Hardness	3.3.1 and GG-I-526	GG-I-526	S-1	*
	Dimensions	3.4	- - -	S-1	4.0
	Surface finish test	3.5	4.4.1	S-1	1.5
	Copper sulfate test (For class 4 material)	GG-I-526	GG-I-526	S-1	*
	Salt-Spray test (For plated parts only)	GG-I-526	GG-I-526	S-1	*

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Table III. Sampling Criteria for Test (Con'td)

Component	Characteristic	Requirement	Test procedure	level	AQL
	Boil test (For all material)	GG-I-526	GG-I-526	S-1	

* Acceptance number shall be zero for all sample sizes.

** Four ounces of each type of each metal used in the construction of the cutter shall be selected for chemical anaylsis.

4.2.3 Sample unit usage. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. Examination shall be conducted as specified in Tables II and IV. Inspection, however, shall not be restricted to defects listed in Table IV. Any bone pin cutters in the sample containing one or more defects shall be rejected and if the number of defective cutters in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

Table IV. Classification of Defects

Categories	Defects
Critical	None defined.
Major	
101	Cutting edges not sharp or free from nicks, burrs, indentations, or other irregularities that affect cutting performance.
102	Cutting edges not properly aligned (i.e., do not meet in same plane) when jaws are closed.
103	Cutting edges do not close completely along entire length.
104	Cutting action not free from excessive stiffness.
105	Cutter not free from cracks, deep pits, burrs or sharp edges.
106	Plating of handles (when applicable) not free from flaking, cracks or blisters.
107	Plating of applicable parts incomplete or not continuous.

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Table IV. Classification of Defects (Con'td)

108	Cutting jaws excessively loose at link pivoting point.
109	All cutting jaw bolts not properly tightened.
110	Cutter does not cut 3/32 inch diameter and 3/16 inch diameter Steinmann and Kirschner bone pins at different positions along cutting [illegible] one cutting stroke
111	Indentations appear on jaw cutting edges after cutting Steinmann and Kirschner bone pins.
Minor	
201	Finish of components, other than cutting jaws, not as specified.
202	Finish of cutter not free from excessive grind marks or excessive roughness.
203	Identification illegible, incomplete, incorrect, or missing.
204	Angle of travel between closed position and maximum opening position of the handles exceeds 120°.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Surface finish test. The following components of the cutter shall be tested for roughness height: the cutting jaws, and top and bottom aligning plates. These components may be tested prior to, or after, final assembly. Should the cutters be assembled, the top and bottom plates shall be disassembled in order to test both surfaces of the cutting jaws and aligning plates. Surfaces shall be tested with an appropriate instrument that complies with requirements specified in section 4 of A.N.S.I. Standard B46.1 and checked with requirements specified in sections 5 and 6 of A.N.S.I. B46.1.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One cutter, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table V shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

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Table V. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	Not Required	12 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Cleaning and drying. Each unit shall be thoroughly cleaned and dried in accordance with the applicable method as outlined in MIL-P-116.

5.1.3.2 Preservative. Not required.

5.1.3.3 Intimate wrap. Each unit shall be wrapped in neutral tissue or other suitable anticorrosive material.

5.1.3.4 Unit package. Each unit shall be packaged in accordance with MIL-P-116, method IA-15.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3. Closure and strapping shall be as specified in the appendix of the applicable box specification.

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5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commerical containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit packages shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The cutter is used to cut Steinmann and Kirschner type bone pins. The triple action design provides the additional leverage necessary to cut this material.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification covers FSN 6515-323-5380 CUTTER, BONE PIN, 18 inch, listed in the Federal Supply Catalog.

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6.4 This specification is intended to cover the types which are normally procured to meet Military requirements. It does not include all types, sizes or grades as indicated by the title nor all types which are commercially available.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-1484

Review activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements.

SPECIFICATION

MIL-C-36676 FSN:

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO (If "yes", in what way?)

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity) | DATE

DD FORM 1426 REPLACES EDITION OF 1 OCT 62 WHICH MAY BE USED.

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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